Lecture 11

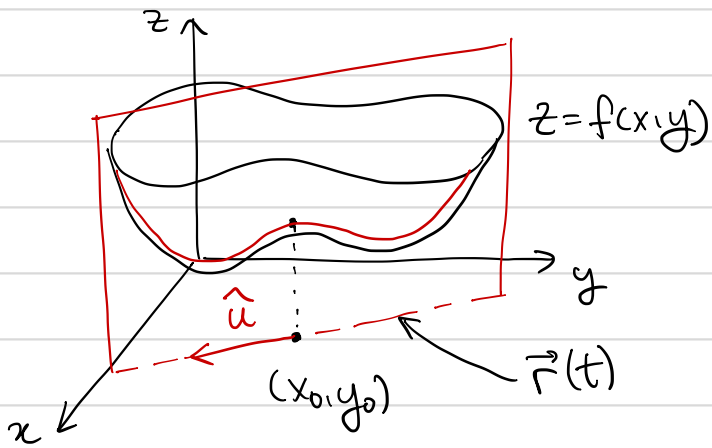
- Gradient vector (14.6)
- Vector fields (16.1)

Directional derivatives

$f = f(x,y) \rightsquigarrow \dfrac{\partial f}{\partial x}, \quad \dfrac{\partial f}{\partial y}$

Direction: $\hat{i}, \quad \hat{j}$

What if I move in another direction $\hat{u}$ = unit vector?



$\hat{u} = \langle a, b \rangle$

$\Rightarrow \vec{r}(t) = (x_0 + at, y_0 + bt)$

Def: $\boxed{D_{\hat{u}} f(x_0, y_0) := \left.\dfrac{d f(\vec{r}(t))}{dt}\right|_{t=0}}$

"Directional derivative"

$D_{\hat{u}} f$ = slope of slice of graph by vertical plane $\parallel \hat{u}$

$$D_{\hat{u}} f = \frac{df}{ds} = f_x \overbrace{\frac{dx}{dt}}^{a} + f_y \overbrace{\frac{dy}{dt}}^{b}$$

$$= \nabla f \cdot \hat{u}$$

$\Rightarrow D_{\hat{u}} f$ = component of ∇f in direction of $\hat{u}$

Example: $D_{\hat{i}} f = \nabla f \cdot \hat{i} = \dfrac{\partial f}{\partial x}$

Meaning of the gradient

$$D_{\hat{u}} f = \nabla f \cdot \hat{u}$$
$$= |\nabla f| \cdot |\hat{u}| \cdot \cos\theta$$
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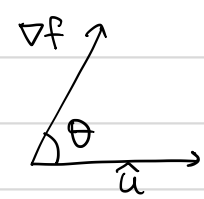


$\hookrightarrow$ max (≥ 0) when $\theta = 0$

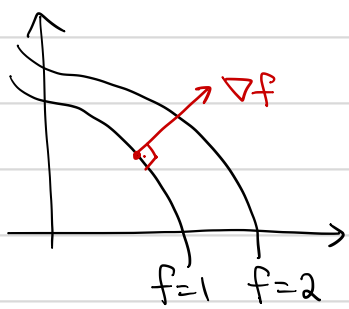
min (≤ 0) when $\theta = \pi$

0 when $\theta = \pi/2$

Direction of ∇f =
= direction along which
f increases the fastest.

Magnitude $|\nabla f|$ =
= Fastest rate of
increase.

<u>Theorem</u>: $\nabla f \perp$ level set
$\{f = \text{constant}\}$



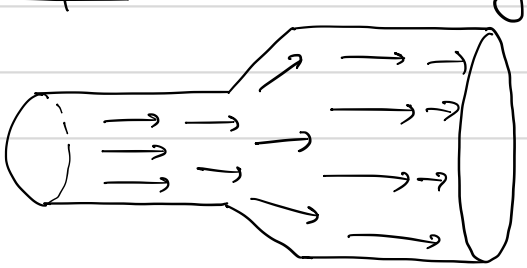
<u>Vector fields</u>

functions

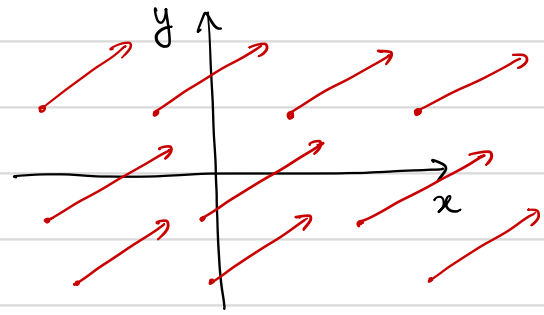
<u>2D</u>: $\vec{F} = \langle P(x,y), Q(x,y) \rangle$
$= P\hat{i} + Q\hat{j}$

<u>3D</u>: $\vec{F} = \langle P(x,y,z), Q(x,y,z), R(x,y,z) \rangle$
$= P\hat{i} + Q\hat{j} + R\hat{k}$

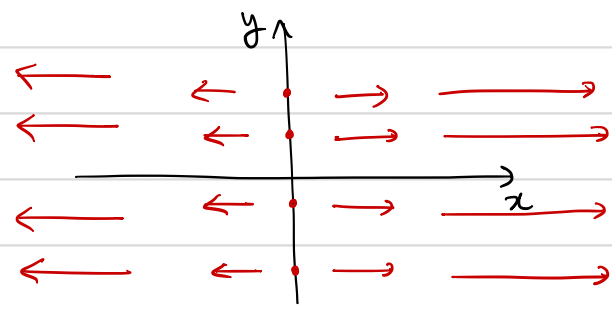
<u>Example</u>: Fluid velocity



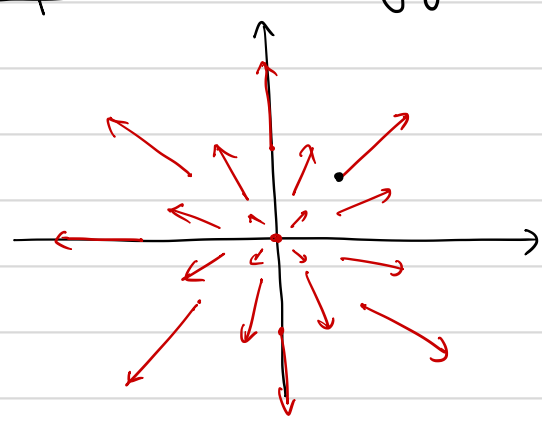
<u>Example</u>: $\vec{F} = 2\hat{i} + \hat{j}$



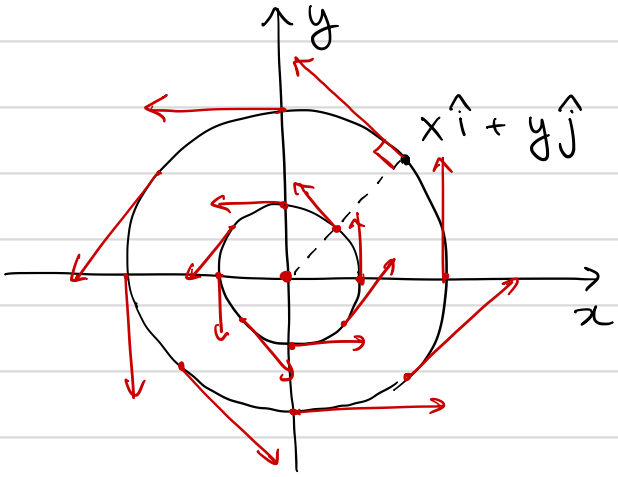
<u>Example</u>: $\vec{F} = x\hat{i}$



<u>Example</u>: $\vec{r} = x\hat{i} + y\hat{j}$

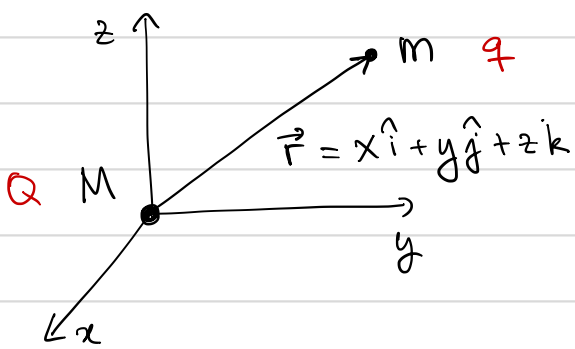


Example: $\vec{F} = -y\,\hat{i} + x\,\hat{j}$



Velocity field of uniform rotation w/ angular speed 1

Example: Newtonian gravity



Force of M on m:

$$|\vec{F}| = \frac{GMm}{|\vec{r}|^2}$$

$$\vec{F} = -\frac{GMm}{|\vec{r}|^3}\,\vec{r}$$

Example: Electrostatics

$$\vec{F} = \frac{\varepsilon_0 Q q}{|\vec{r}|^3}\,\vec{r}$$

Example: Gradient fields

$$\vec{F} = \nabla f = \langle f_x, f_y, f_z \rangle$$